

Lecture 01 - Introduction

This lecture is the orientation layer for the whole course. It explains why interactive computer graphics is a pipeline problem: a scene must be represented, processed under time constraints, sampled into an image, and displayed fast enough for interaction.

Source chunk	course_text_parts/03_lectures/01_introduction.txt
Extracted slide pages	51
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Slide walkthrough

Each row gives a short reading comment for one extracted slide page. Use it to turn sparse slide bullets into a human-readable path through the lecture.

Page	Slide title	How to read it
1	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
2	Course Learning Outcomes	Read this as organization and relevance information: it tells you how lecture theory, exercises, and exam preparation connect.
3	Real-Time Rendering Contest Results 1/3	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
4	Real-Time Rendering Contest Results 2/3	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
5	Real-Time Rendering Contest Results 3/3	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
6	1.1 Course Organization	Read this as organization and relevance information: it tells you how lecture theory, exercises, and exam preparation connect.
7	Teachers	Read this as organization and relevance information: it tells you how lecture theory, exercises, and exam preparation connect.
8	Course Format	Read this as organization and relevance information: it tells you how lecture theory, exercises, and exam preparation connect.

Page	Slide title	How to read it
9	Course Format	Read this as organization and relevance information: it tells you how lecture theory, exercises, and exam preparation connect.
10	Exercises	Read this as organization and relevance information: it tells you how lecture theory, exercises, and exam preparation connect.
11	Successful Course Completion	Read this as organization and relevance information: it tells you how lecture theory, exercises, and exam preparation connect.
12	Course Overview	Read this as organization and relevance information: it tells you how lecture theory, exercises, and exam preparation connect.
13	Literature	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
14	Relevant Chapters and Link to Lecture Course	Read this as organization and relevance information: it tells you how lecture theory, exercises, and exam preparation connect.
15	(1.1 Course Organization)	Read this as organization and relevance information: it tells you how lecture theory, exercises, and exam preparation connect.
16	1.2 Computer Graphics Overview	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
17	Computer Graphics History	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
18	Computer Graphics History	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
19	Computer Graphics History	Read this as boundary logic. Identify what is inside, what is outside, and when new intersection vertices are created.
20	Computer Graphics History	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
21	Computer Graphics History	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
22	Related Disciplines	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
23	Applications	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?

Page	Slide title	How to read it
24	1.3 Pixel-based Representations	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
25	Raster Images	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
26	Pixel	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
27	Color	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
28	LCD Raster Screen	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
29	Raster Image Sizes	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
30	1.4 3D Models	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
31	3D Model Types	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
32	Implicit Surfaces	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
33	Quadrics	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
34	Object Representation	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
35	Polygonal Models 1/3	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
36	Polygonal Models 1/4	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
37	Polygonal Models 2/4	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
38	Polygonal Models 3/4	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.

Page	Slide title	How to read it
39	Polygonal Models 4/4	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
40	Volumetric Models	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
41	Model Generation	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
42	Coordinate Systems	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
43	Rendering	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
44	1.5 Algorithmic Paradigms	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
45	Photorealistic Rendering	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
46	Non-Photorealistic Rendering (NPR)	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
47	Image-Based Algorithms	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
48	Object-Based Algorithms	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
49	are assigned	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
50	Literature and other sources used in this chapter	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
51	Peters 2016. (Chapters 1, 3 & 21)	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.

01.1 Course Organization

Source location: Slides around 6-15

Commentary

The organizational slides matter because they explain how the theory and programming parts fit together. The exercises are not separate from the lecture; they are the practical version of the same pipeline ideas. When the course says that slides are generally not self-explanatory, that is a warning: the bullet points are anchors, not a textbook.

Treat every exercise sheet as an applied checkpoint. Rendering pipeline, primitive types, shaders, transformations, projection, clipping, and rasterization are introduced in the lectures and then turned into OpenGL tasks. If a topic appears both in a lecture and an exercise, it is more likely to be exam relevant.

Mental model

Lecture slides give the vocabulary; exercises force you to connect that vocabulary to code and visible output.

Check yourself

Can you list the five exercise themes and connect each one to a later lecture chapter?

01.2 Computer Graphics Overview

Source location: Section 1.2

Commentary

Computer graphics is about generating images from descriptions. The description can be pixel-based, object-based, physically motivated, artistic, or algorithmic. The common thread is that the computer must decide what color belongs at image samples.

Interactive graphics adds a time constraint. The image is not computed once and admired; it must respond to camera motion, user input, animation, and changing state. That is why real-time rendering often uses approximations and specialized GPU stages.

Mental model

Graphics is controlled image generation. Interactive graphics is controlled image generation under a strict time budget.

Check yourself

What changes when a renderer must be interactive rather than offline?

01.3 Pixel-Based Representations

Source location: Section 1.3

Commentary

A pixel-based representation stores an image directly as samples on a grid. It is easy to display and edit locally, but it does not know the underlying 3D scene. If you zoom into a raster image, you reveal the sampling grid, not more geometry.

This is the first place where sampling becomes important. Resolution, color depth, aliasing, and frame time are not cosmetic details. They decide what information can be represented and how much data must be processed per second.

Mental model

A raster image is already the answer. It contains color samples, not the geometric reasons behind them.

Check yourself

Why can a pixel image be easy to display but hard to reinterpret as a 3D scene?

01.4 3D Models

Source location: Section 1.4

Commentary

A 3D model is a structured description of objects before they become pixels. It may contain vertices, primitives, topology, normals, materials, textures, and transformations. Unlike a pixel image, it can be viewed from different camera positions.

The course mostly follows object-based rendering: start from model data, transform it, project it, rasterize it, shade it, test visibility, and finally update pixels. Later chapters explain these steps one by one.

Mental model

A model is not an image. It is a cause from which many possible images can be generated.

Check yourself

Which model attributes can influence final color besides vertex positions?

01.5 Algorithmic Paradigms

Source location: Section 1.5

Commentary

The lecture contrasts ways to generate images. Rasterization works from objects toward pixels and is the dominant real-time pipeline. Ray-based methods work from image samples into the scene and are powerful for visibility and lighting but can be more expensive.

Do not memorize these paradigms as names only. Ask what direction information flows. Does the algorithm start from geometry and find covered pixels, or from pixels/rays and find visible geometry? That direction explains the strengths and weaknesses.

Mental model

Rasterization asks, 'Which samples does this primitive cover?' Ray casting asks, 'What does this sample see?'

Check yourself

Which paradigm fits OpenGL's standard real-time pipeline most directly?

End-of-lecture summary

If you remember only one thing from Lecture 01, remember this: This lecture is the orientation layer for the whole course. It explains why interactive computer graphics is a pipeline problem: a scene must be represented, processed under time constraints, sampled into an image, and displayed fast enough for interaction.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.